

RYONGYON, Korea, Democratic People's Republic of

WMO: 470680

Lat: 38.150N

Lon: 124.883E

Elev: 6

StdP: 101.25

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.7	-10.7	-17.6	0.8	-9.0	-15.8	0.9	-8.0	9.4	-5.4	8.8	-5.1	2.0	320	0.421

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.2	31.0	25.5	29.7	24.7	28.6	24.0	26.7	29.8	26.0	28.6	25.3	27.6	2.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.7	21.0	28.7	25.2	20.3	27.8	24.6	19.6	27.1	83.8	30.1	80.6	28.8	77.8	27.7	32.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.9	7.8	6.7	-16.2	32.8	2.9	1.0	-18.3	33.5	-20.0	34.1	-21.6	34.7	-23.7	35.4		
			-16.2	28.0	2.6	1.3	-18.0	28.9	-19.6	29.7	-21.0	30.5	-22.9	31.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.3	-3.7	-1.3	4.1	10.6	16.3	21.2	24.4	24.9	20.0	13.6	6.0	-1.1
	DBStd	10.40	3.82	3.72	3.28	3.27	2.51	2.02	1.91	2.23	2.58	3.29	4.11	4.04
	HDD10.0	1441	424	316	186	32	0	0	0	0	0	8	131	343
	HDD18.3	3118	683	549	443	233	72	3	0	0	13	150	371	601
	CDD10.0	1918	0	0	2	49	196	336	447	461	299	119	10	0
	CDD18.3	556	0	0	0	0	9	89	189	203	63	3	0	0
	CDH23.3	3731	0	0	0	3	97	472	1226	1590	329	13	0	0
	CDH26.7	831	0	0	0	0	8	82	273	431	36	1	0	0
Wind	WSAvg	2.3	2.1	2.4	2.8	2.9	2.7	2.1	2.2	1.9	1.5	1.9	2.3	2.3
Precipitation	PrecAvg	864	12	18	19	50	80	94	243	185	91	36	38	21
	PrecMax	1685	33	62	52	111	253	276	508	446	431	100	123	47
	PrecMin	447	1	0	0	3	4	12	45	29	5	1	4	3
	PrecStd	257	9	15	14	26	55	63	121	114	97	27	29	13
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.5	10.7	16.6	23.3	27.8	30.4	32.5	33.0	29.3	25.0	18.2	10.7
		MCWB	5.2	7.3	10.0	15.1	18.6	21.3	26.8	26.7	22.9	17.7	14.5	8.3
	2%	DB	5.3	7.8	13.8	20.5	25.8	28.8	30.6	31.5	27.7	22.8	16.1	8.6
		MCWB	2.9	4.7	8.7	13.7	17.1	20.9	25.6	26.2	21.6	17.1	13.1	6.4
	5%	DB	3.6	6.2	11.8	18.7	24.1	27.4	29.3	30.2	26.5	21.2	14.5	6.7
		MCWB	1.5	3.4	7.5	12.3	16.4	20.6	25.0	25.6	20.9	16.0	11.7	4.5
	10%	DB	2.1	4.6	10.0	16.9	22.4	26.1	28.1	29.0	25.4	19.7	12.8	5.0
		MCWB	0.2	2.4	6.2	11.3	15.7	20.1	24.5	25.1	20.3	15.3	10.1	3.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.6	8.2	11.5	17.1	20.1	23.4	27.7	27.7	24.9	19.6	15.6	9.0
		MCDB	7.1	9.9	14.4	20.8	24.9	27.1	30.9	31.2	27.0	22.7	17.1	10.6
	2%	WB	3.2	5.4	9.7	15.1	18.7	22.5	26.6	26.9	23.2	18.2	13.7	6.6
		MCDB	4.8	7.3	12.6	19.0	23.4	25.9	29.6	30.2	25.9	21.3	15.3	8.1
	5%	WB	1.7	3.8	8.1	13.6	17.7	21.8	25.8	26.2	22.2	17.1	12.0	4.7
		MCDB	3.3	5.7	11.3	17.2	21.9	25.3	28.2	29.1	25.2	20.0	14.2	6.4
	10%	WB	0.4	2.5	6.6	12.2	16.8	21.2	25.1	25.7	21.3	16.0	10.3	3.2
		MCDB	1.9	4.4	9.5	15.7	20.7	24.5	27.2	28.2	24.3	18.9	12.4	4.9
Mean Daily Temperature Range	5% DB	MDBR	8.0	8.4	9.1	10.3	10.7	8.9	6.2	7.2	10.0	10.8	8.5	7.5
		MCDBR	9.8	10.9	12.4	13.2	13.9	11.7	8.5	8.7	11.4	12.8	11.1	9.6
		MCWBR	7.9	8.1	8.0	7.4	6.7	5.2	4.5	4.3	5.9	7.3	8.0	7.6
	5% WB	MCDBR	9.2	9.5	11.1	10.7	11.0	8.4	6.6	7.1	9.0	10.8	9.8	9.0
		MCWBR	7.7	7.7	8.2	7.0	6.5	4.5	4.1	4.1	5.5	7.2	8.0	7.6
Clear-Sky Solar Irradiance	taub	0.438	0.496	0.557	0.613	0.630	0.714	0.686	0.590	0.519	0.531	0.510	0.447	
	taud	1.938	1.828	1.730	1.664	1.680	1.588	1.691	1.902	1.998	1.849	1.849	1.927	
	Ebn at Noon	686	698	702	694	693	636	650	705	730	657	604	639	
	Edh at Noon	139	176	214	242	243	266	238	189	162	170	151	132	
All-Sky Solar Radiation	RadAvg	2.26	3.16	4.38	5.06	5.58	5.12	4.32	4.64	4.44	3.55	2.35	1.96	
	RadStd	0.18	0.24	0.36	0.35	0.52	0.59	0.58	0.56	0.38	0.23	0.26	0.13	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page